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1a. Kimball four-stage documentation

1.1 Stage 1 – Business Process: Identify the primary business process being modelled and explain why it is the most appropriate choice given the data available.

The primary business process is retail sales transactions because FreshMart's main activity is selling products to customers. This is the most appropriate choice because Kimball and Ross (2013) state that a business process should be a specific operational activity that produces measurable metrics. And, the other data, such as product, store, customer, and promotion, provide descriptive context to support the sale transaction.

1.2 Stage 2 – Grain: Precisely declare the grain of the fact table. Explain why you chose this grain over alternatives.

The grain of the fact table is that each row represents the sale of a specific product to a particular customer at a particular store on a given date under a particular promotion. The main benefit of using a detailed grain is that it keeps all the details in the data, so in the future, if the requirements change or other team members want to analyse it after we leave, they can still do so. For example, if one day FreshMart wants to analyse which products are often bought together in the same transaction, this would only be possible if the data is kept at a detailed grain.

Transaction-level grain: Each individual product sold in a sales transaction is recorded as a separate row.

Alternative aggregation option: Sales could be aggregated by day, meaning multiple product sales in the same store on the same day would be combined into a single row.

1.3 Stage 3 – Dimensions: List all dimensions and briefly justify their inclusion.

The dimensions included in the model are Product, Customer, Store, and Promotion. According to Kimball and Ross (2013), dimensions describe the business event by providing the “who, what, where, when, why, and how” that is related with each measurement. The Product dimension is necessary to analyse which product categories and subcategories drive the most revenue. The Customer dimension helps the business analyse customer behaviour and buying patterns. For example, which group of customers tend to spend more? The Store dimension is used to compare sales performance across stores and regions. The Promotion dimension allows businesses to analyse the effectiveness of promotions on sales performance. The Date dimension was created to enable time-based analysis, such as YTD, PYTD, and YoY%, which we need for this assessment. While the date in the Sale table just shows when the transaction happened, the Date dimensions will work as a complete calendar to support year-to-date or prior year analysis.

1.4 Stage 4 – Facts: Identify all additive, semi-additive, and non-additive facts and classify them accordingly.

In the Sales fact table, the measures include QuantitySold, UnitPrice, DiscountAmount, and COGS. Among these, QuantitySold, DiscountAmount, and COGS are additive facts because they can be summed meaningfully across all dimensions. UnitPrice is a non-additive fact

because it cannot be summed directly and must be used in calculations. There are no semi-additive facts in this model because the dataset is transaction-level sales data, not balance or snapshot data.

1b. Entity-Relationship / Star Schema Diagram

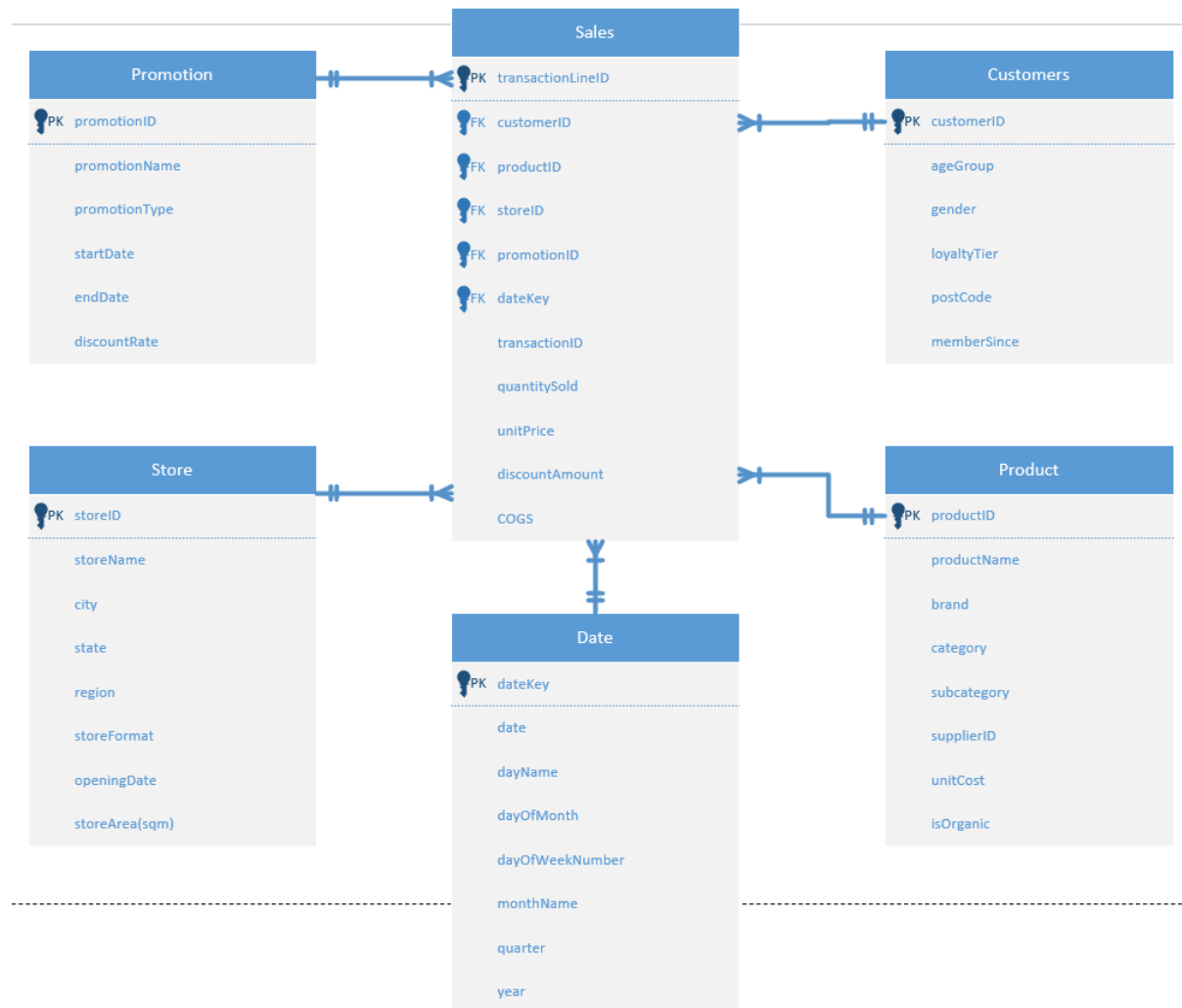


Figure 1. Entity-relationship / star schema diagram for the FreshMart data model (created by the author).

2a. Data quality audit table

Table 1. Data quality audit summary for the FreshMart source tables

Table	Column	Problem	Rows	Remediation action
Customer	CustomerID, MemberSince	Duplicate customer records were identified with the same CustomerID but slightly different membership dates	30	Removed duplicate rows based on CustomerID in Power Query
Product	Category	Inconsistent spelling and capitalisation in category values, such as dairy, Diary, and bakery	8	Standardised category labels in Power Query using a cleaned column and value replacement
Promotion	PromotionType, StartDate, EndDate	N/A was used in PromotionType for the no-promotion record, and blank values were found in StartDate and EndDate	1	Replaced N/A with No Promotion to make it more business-friendly. The blank values in StartDate and EndDate were kept as null because entering No Promotion or N/A would change the columns to text data type
Sales	Date	Mixed date formats were found in the raw data when viewed in Notepad	-	Power BI changes the mixed date formats automatically during import, so the main action was to confirm that the column type was set correctly as Date
Sales	CustomerID	The value C9999 did not match any record in the Customer table	93	Create the CustomerID_Clean column to replace C9999 from CustomerID with "Not Applicable" in the copied column for clearer handling of unmatched customer records
Store	SquareMetres	The column name SquareMetres was	-	Renamed SquareMetres to

		less user-friendly for reporting purposes		Store Area (sqm)
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Sources: Created by the author from the FreshMart source tables provided in the assignment

Model view screenshot

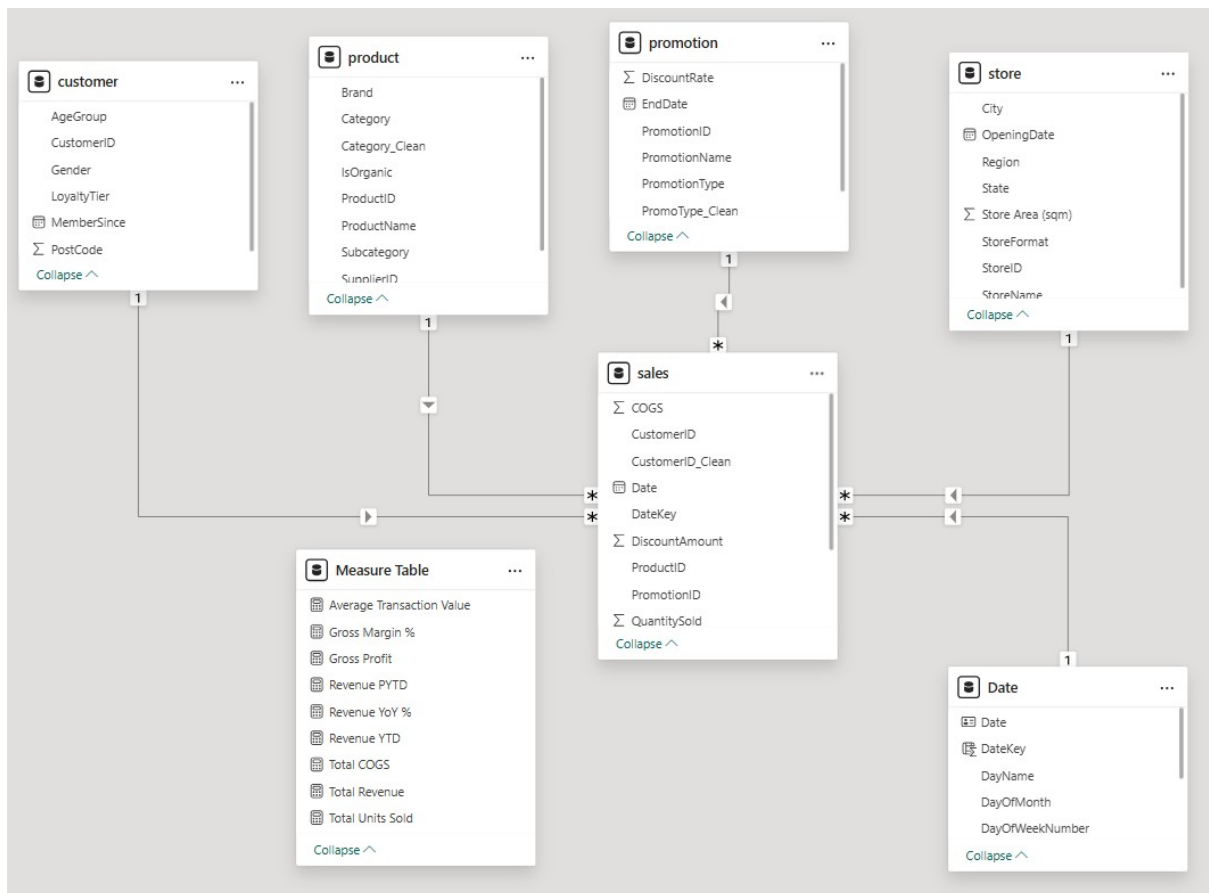


Figure 2. Power BI model view for the FreshMart star schema.

Appendix: Applied Steps screenshots for each table

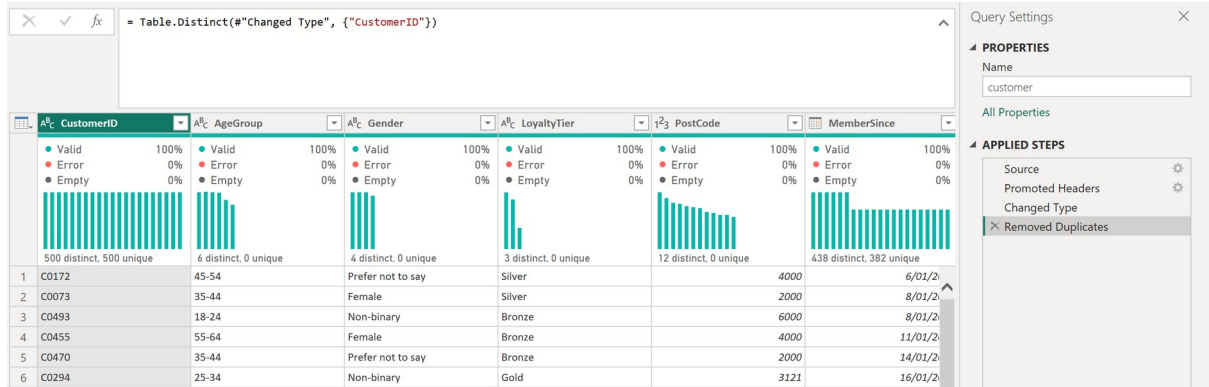


Figure 3. Applied Steps screenshot for the Customer table in Power Query

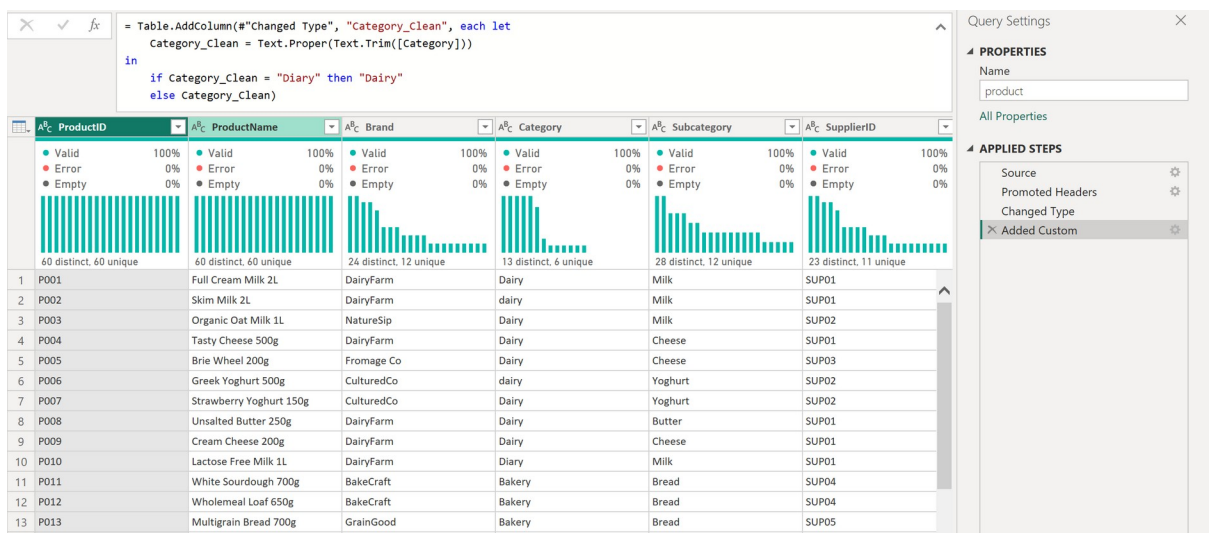


Figure 4. Applied Steps screenshot for the Customer table in Power Query

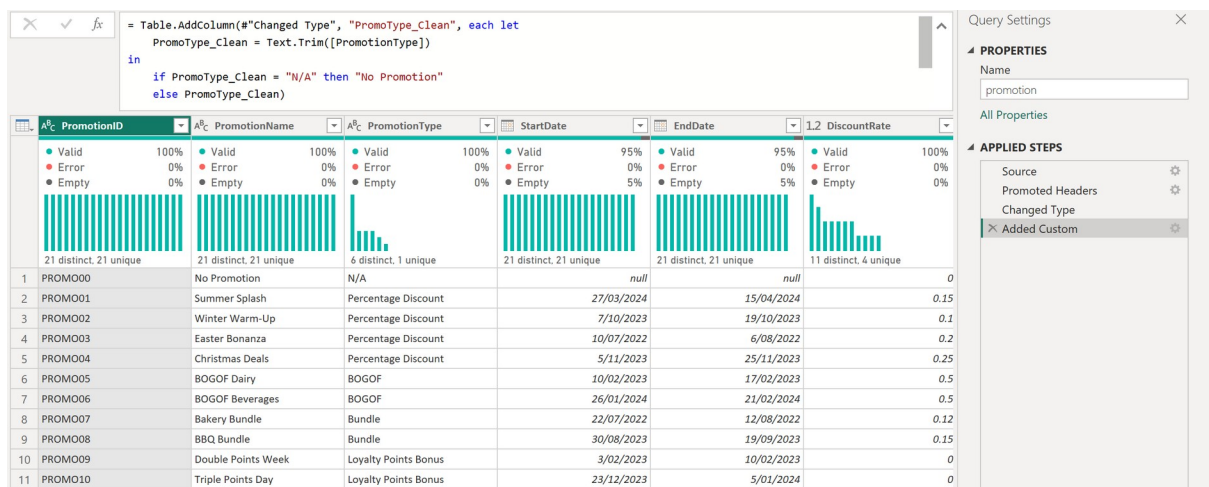


Figure 5. Applied Steps screenshot for the Product table in Power Query

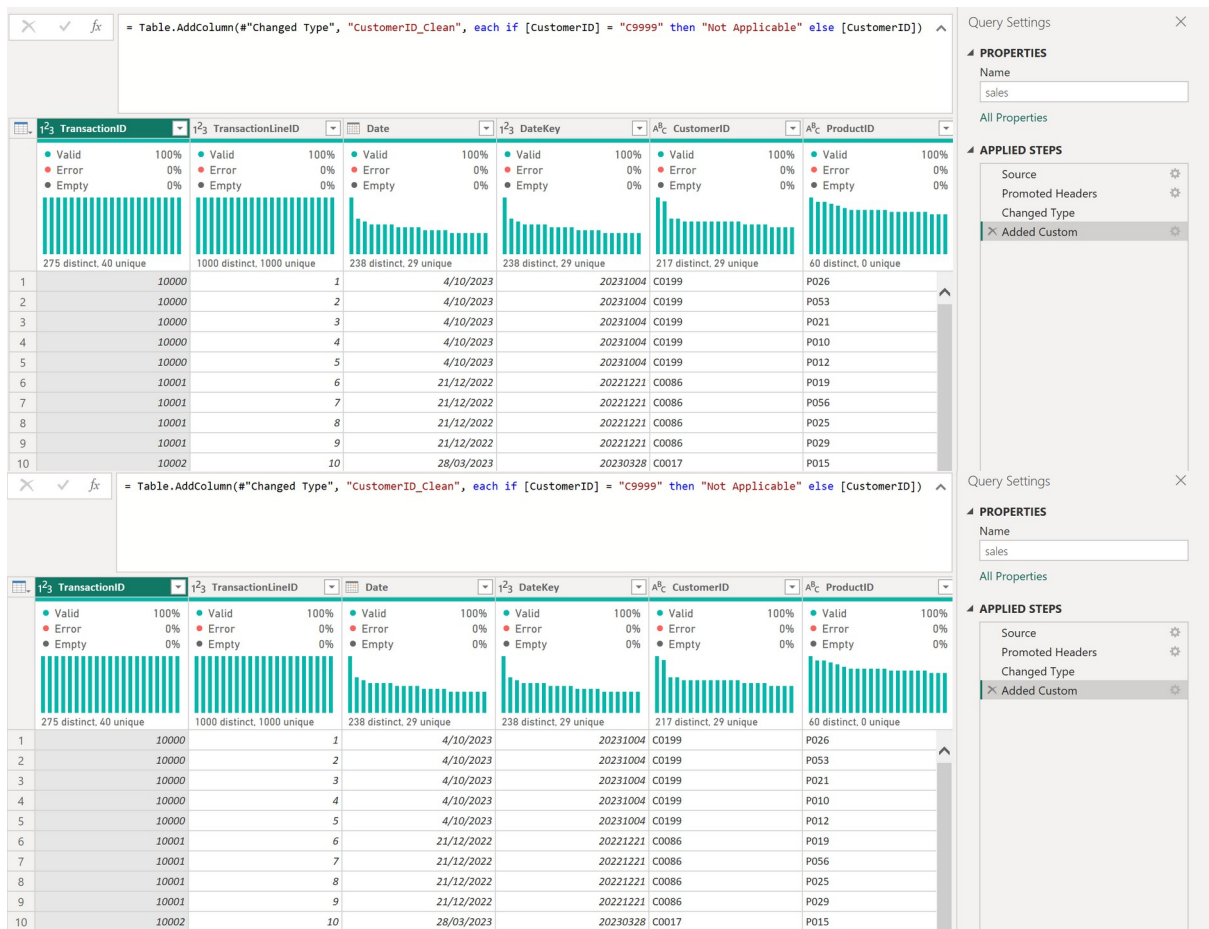


Figure 6. Applied Steps screenshot for the Sales table in Power Query

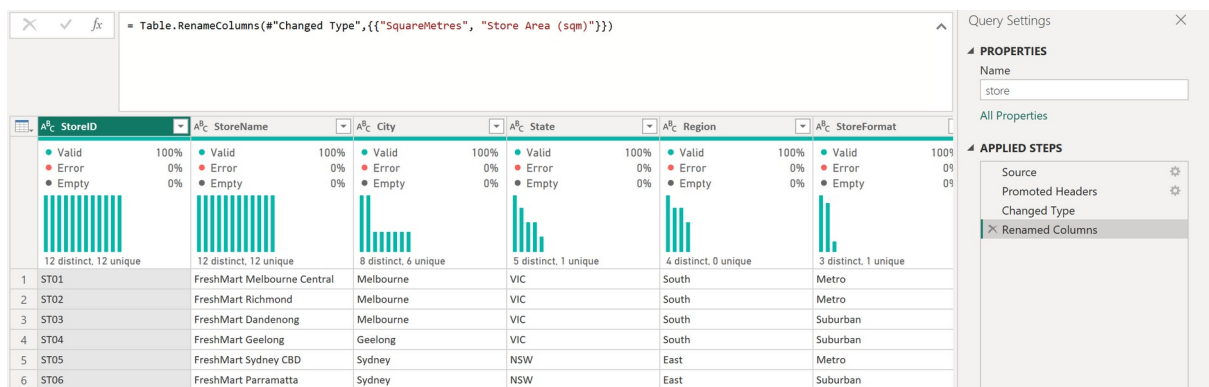


Figure 7. Applied Steps screenshot for the Store table in Power Query

Reference List

Kimball, Ralph, and Margy Ross. 2013. *The Data Warehouse Toolkit : The Definitive Guide to Dimensional Modeling*. John Wiley & Sons, Incorporated.